

AMATH 732

Quiz 2; 30 Minutes

Question 1: 5 marks Consider the variable mass linear oscillator

$$(m_0 + \Delta m \sin(t)) \frac{d^2 x}{dt^2} + kx = 0$$

with the initial conditions $x(0) = x_0$ $v(0) = 0$. If the idea is that Δm is a small change, non-dimensionalize the problem and define a dimensionless small parameter ϵ .

Question 2: 5 marks Given the undamped Duffing oscillator (where the assumption is that nonlinearity matters, but is not dominant)

$$m \frac{d^2 x}{dt^2} + kx + k_3 x^3 = 0$$

with the initial conditions $x(0) = x_0$ and $v(0) = v_0$ find the dimensionless form of the equation. Define all terms you use.